


www.advanceddevices.com

Feedback - Adre hv G

ELECTRONS.PSYCHOGENIC.COM

[NEWS](#)[RAVES & RANTS](#)[GAMES](#)[CONTACT US](#)[HOME](#)[ACCOUNT](#)[PRIVATE MESSAGE](#)[Main Menu](#)[Home](#)[Latest News](#)[Raves & Rants](#)[Articles](#)[Add Article](#)[My Articles](#)[Newest](#)[Top Rated](#)[Most Popular](#)[Frozen Bubble](#)[Flash Games](#)[Login](#)

Username:

Password:

[User Login](#)[Secure Login](#)[Lost Password?](#)[Register now!](#)**Electrons :: Articles :: AVR :: AVR In-System Programmer**

AVR

AVR In-System Programmer

Description: AVR In-System Programmer. Here we provide details on our version of the Atmel AVR In-System Programmer (ISP). We provide schematics and printed circuit board (PCB) art to allow you to construct your own programmer.

1. AVR In-System Programmer**AVR In-System Programmer**

Copyright (C) 2003,2004, Psychogenic Inc. All Rights Reserved.

Here we provide details on our version of the Atmel AVR In-System Programmer (ISP). We provide schematics and printed circuit board (PCB) art to allow you to construct your own programmer.

TOC

- [Psychogenic PPI Programmer](#)
 - [Psychogenic PPI Programmer Schematic](#)
 - [PCB](#)
 - [PCB layout](#)
 - [Parts List](#)
- [PsychoISP Programmer](#)
 - [Psychogenic AVR910 ISP Schematic](#)
 - [PCB](#)
 - [PCB layout](#)
 - [Parts List](#)
 - [Programmed uC](#)

This document details the construction of Psychogenic's two versions of an AVR ISP programmer--it's focus is on the schematics, components and construction of the programmers. A good deal of details on the actual programming process is available in our [AVR Programmer HOWTO](#) page.

As mentioned in our [AVR Programmer](#) section, a quick and easy solution for beginners is the the parallel port programmer described on the [avrdude](#) site. It uses nothing more than a 25 pin parallel port connector, ceramic resonator and a ZIF socket.

Here we present two distinct programmers. Both are used to program AVR chips using the [low voltage serial programming](#) interface. The first uses AVR Dude's parallel port interface ([PPI](#)) while the second uses the computer's serial port in the standard [AVR910](#) configuration.

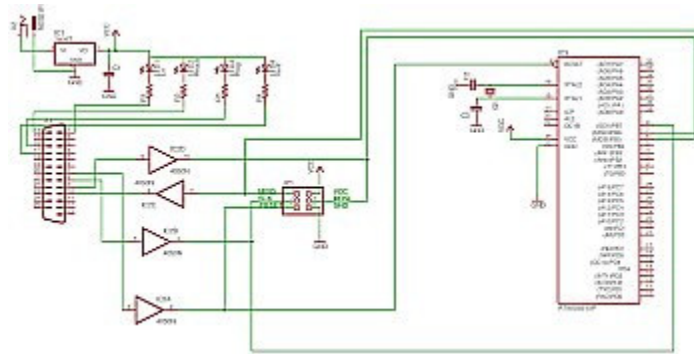
Psychogenic PPI AVR Programmer

In this section, we will detail the construction of our PPI AVR Programmer. It connects to your computer's parallel port and implements a good deal of `avrdude`'s features (e.g. status LEDs). Features we've added include an independent power source and voltage regulator (to protect your microcontroller *and* parallel port/motherboard), a 40 pin socket to allow you to program AT90S8515 chips directly as well as a standard 6 pin ISP header so you can program all other serial programmable chips, in circuit.

Downsides of this programmer are the fact it uses the parallel port (and thus requires a more expensive 25-wire cable) and only provides the 6 pin ISP header.

Psychogenic PPI Programmer Schematic

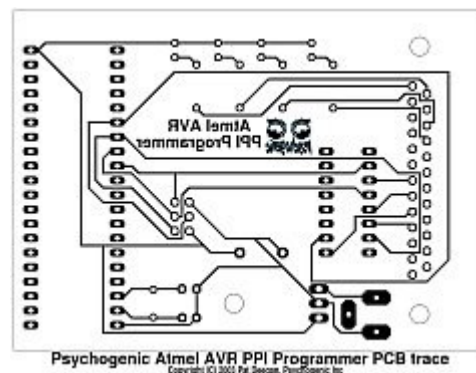
Here is the schematic for our PPI programmer:



Click the image for a [large view of the schematic](#). This programmer uses an independent power supply, regulated down to +5V, and adds a buffer between the parallel port and microcontroller. It also sports a number of LEDs for visual feedback and the standard Atmel 6 pin ISP header, to program all types of AVR microcontrollers in circuit.

PPI PCB

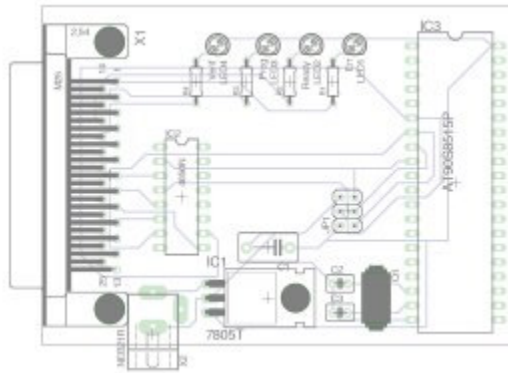
You may use our PCB artwork (according to the terms set forth in our [license](#)) to [create](#) : [sturdy and professional printed circuit board](#).



Do not use the image above (it is low-res and won't be quite to scale), download instead a [high-resolution \(400 dpi\) PCB](#) and have it printed at **400 dpi**. Be sure to check our PCB creation technique and hints on our [Printed Circuit Board \(PCB\) HOWTO](#).

PPI Component Placement

Once you've created your PCB, use this diagram and the parts list below to complete the PPI Programmer circuit.

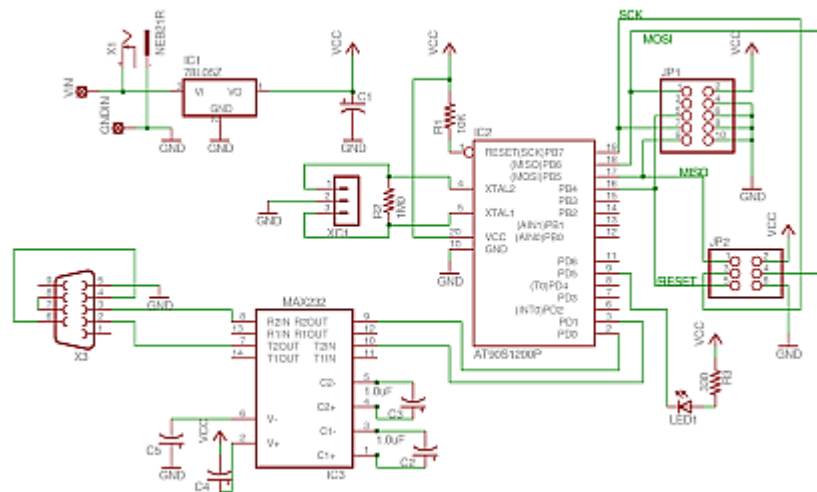


PsychoISP: AVR910 ISP programmer

This is our version of the [AVR910](#) serial programmer.

Psychogenic AVR910 ISP Schematic

A slightly more sophisticated home-brew programmer would be to build our solution, a serial ISP programmer with both 6- and 10-pin header outputs based on the programme described in the [AVR910: In-System Programming](#) app note:



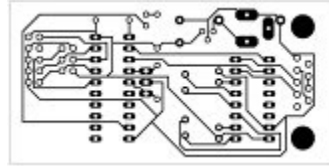
Click on the image for a [larger view of the schematic](#). Circuit features:

- Separate, regulated, power supply
- Both standard ISP headers: 6 pin and 10 pin
- Activity LED indicator
- Use of standard RS-232 interface allows for serial cable up to 50 feet long
- Available printed circuit board (see below)

Downsides include

PCB

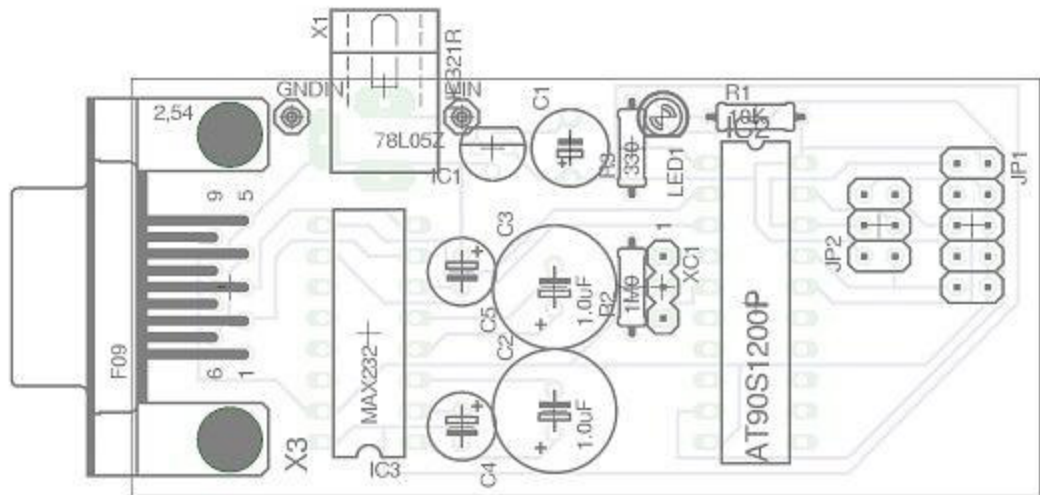
A very clean way to produce this circuit is to use our printed circuit board (you are free to use the image to create your own PCB trace, for non-commercial purposes, see our [license](#) for details):



Do not use the above image as it will not be the exact size required, click instead on the image to download a 400dpi (hi-res) version of the PCB. Print that image at 400 dpi to get a "life size" inverted image of the PCB that can easily be transferred to a copper board for etching. If you are new to PCB creation (or want to see if you can learn a few new tricks), have a look at our [PCB Creation Tutorial](#). As always, be careful when playing with the outside world.

PCB layout

Once your PCB has been etched and drilled (read the [disclaimer](#)), components should be placed as in the following image. Make sure to note the orientation of the ICs, polarized capacitors, LED and other components.



Parts List

Part	Value	Description
C1	1.0 uF	Capacitor
C2	1.0 uF	Capacitor
C3	1.0 uF	Capacitor
C4	1.0 uF	Capacitor
C5	1.0 uF	Capacitor
IC1	78L05Z	5V voltage regulator
IC2	AT90S1200P	Atmel AVR uC

IC3	MAX232	RS-232 interface
JP1	10 pin header (2x5)	10 pin ISP
JP2	6 pin header (2x3)	6 pin ISP
LED1	3MM LED	Activity LED
R1	10k ohm	Resistor
R2	1Meg ohm	Resistor
R3	330 ohm	Resistor
X1	power jack	2.1mm power jack
X3	F09H	Female 9pin sub-d conn
XC1	Ceramic resonator	4MHz freq resonator
VIN	Voltage In	Power supply input pad
GNDIN	GND In	Power ground input pad

Programmed uC

Although this ISP programmer is a great solution it's main issue is *bootstrapping*: you need a programmed AT90S1200 chip in order to use it to program other microcontrollers... To break this circular dependency, you can use a temporary construction of the parallel port programmer from the [avrdude site](#) or a friend's programmer to write a compiled version of this [AVR910.ASM](#) file (a slightly modified version of the Atmel ISP programmer code) to an AT90S1200 microcontroller or you can [order a pre-programmed device](#) from us.

Level: [Article](#)

Additional Article Data

Level: [Article](#)

Comments

[Nested](#) ☐ [Oldest First](#) ☐ [Refresh](#)

The comments are owned by the poster. We aren't responsible for their content.

Jump to section

[AVR](#) ☐ [Go](#)



AVR

[<< Prev](#) | [Ring Hub](#) | [Join](#) | [Next >>](#)

© 2007 [WebRing Inc.](#)

[Search](#)

All contents are Copyright (C) 2004-2005 [Psychogenic Inc](#) -- All rights reserved